



CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS

DATE: 27/07/2026

EXPERIMENT 11-- Database with VIEW and INDEX

AIM:

To create and demonstrate an **AUTO_INCREMENT sequence** for generating register numbers in a database table.

PROCEDURE:

1. Open MySQL Workbench and select the required database.
2. Use the existing student_cgpa table containing student details and manually entered register numbers.
3. Create a new table student_cgpa_auto with reg_no defined as AUTO_INCREMENT.
4. Insert student records without specifying the register number.
5. Display the table and observe that MySQL automatically generates sequential register numbers.
6. Manually populate register numbers in the existing student_cgpa table.
7. Remove the AUTO_INCREMENT property from student_cgpa_auto using ALTER TABLE.
8. Display the table structure using DESC to verify the changes.

PROGRAM:

1. CREATE TABLE

```
CREATE TABLE student_cgpa (  
    reg_no VARCHAR(15) PRIMARY KEY,  
    student_name VARCHAR(50) NOT NULL,  
    department VARCHAR(20),  
    no_of_subs_pass INT,  
    no_of_subs_fail INT,  
    cgpa DECIMAL(3,2)  
);
```

Then Insert Values:

```
INSERT INTO student_cgpa  
(reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa)  
VALUES  
(192411123, 'Deepthi', 'CSE', 22, 2, 8.71),  
(192512121, 'Varun', 'IT', 27, 1, 9.23),  
(192312117, 'Aadhya', 'ECE', 25, 3, 8.97),
```

```
( '192411227','Dimple','CSE',31,0,9.37),
( '192820896','Mohan','EEE',18,9,6.48),
( '192411122','Yogitha','CSE',31,1,9.01),
( '192516871','Sadhya','IT',25,5,7.22),
( '192111101','Rahul','MECH',30,12,5.14),
( '192411010','Alekhya','CSE',35,2,9.01),
( '192511135','Rahul','CIVIL',37,0,9.72);
```

```
SELECT * FROM student_cgpa;
```

OUTPUT:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192111101	Rahul	MECH	30	12	5.14
192312117	Aadhya	ECE	25	3	8.97
192411010	Alekhya	CSE	35	2	9.01
192411122	Yogitha	CSE	31	1	9.01
192411123	Deepthi	CSE	22	2	8.71
192411227	Dimple	CSE	31	0	9.37
192511135	Rahul	CIVIL	37	0	9.72
192512121	Varun	IT	27	1	9.23
192516871	Sadhya	IT	25	5	7.22
192820896	Mohan	EEE	18	9	6.48

#	Time	Action	Message	Duration / Fetch
1	22:41:47	CREATE TABLE student_cgpa (reg_no VARCHAR(15) PRIMARY KEY, student_name VARCHAR(50) NO...	0 row(s) affected	0.031 sec
2	22:44:31	INSERT INTO student_cgpa (reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa) VA...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
3	22:46:18	SELECT * FROM student_cgpa LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

2. Create AUTO_INCREMENT Table

```
CREATE TABLE student_cgpa_auto (
reg_no INT AUTO_INCREMENT PRIMARY KEY,
student_name VARCHAR(50),
department VARCHAR(20),
no_of_subs_pass INT,
no_of_subs_fail INT,
cgpa DECIMAL(3,2)
);
```

OUTPUT:

#	Time	Action	Message	Duration / Fetch
1	21:33:08	SHOW TABLES	6 row(s) returned	0.000 sec / 0.000 sec
2	21:33:57	SELECT * FROM student_cgpa LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
3	21:34:20	CREATE TABLE student_cgpa_auto (reg_no INT AUTO_INCREMENT PRIMARY KEY, student_name VA...	0 row(s) affected	0.047 sec

3. Populate Register Number Using AUTO_INCREMENT

```
INSERT INTO student_cgpa_auto
(student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa)
VALUES
('Deepthi','CSE',22,2,8.71),
('Varun','CSE',27,1,9.23),
('Aadhya','ECE',25,3,8.97),
('Dimple','CSE',31,0,9.37),
```

```
('Mohan','EEE',18,9,6.48);
SELECT * FROM student_cgpa_auto;
```

OUTPUT:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
1	Deepthi	CSE	22	2	8.71
2	Varun	CSE	27	1	9.23
3	Aadhya	ECE	25	3	8.97
4	Dimple	CSE	31	0	9.37
5	Mohan	EEE	18	9	6.48
NULL	NULL	NULL	NULL	NULL	NULL

4. Manually Populate Register Number

existing student_cgpa table already demonstrates manual register number entry.

```
SELECT reg_no, student_name, department
FROM student_cgpa;
```

OUTPUT:

reg_no	student_name	department
192111101	Rahul	MECH
192312117	Aadhya	ECE
192411010	Alekhyia	CSE
192411122	Yogitha	CSE
192411123	Deepthi	CSE
192411227	Dimple	CSE
192511135	Rahul	CIVIL
192512121	Varun	IT
192516871	Sadhya	IT
192820896	Mohan	EEE
NULL	NULL	NULL

5. Drop the AUTO_INCREMENT

```
ALTER TABLE student_cgpa_auto  
MODIFY reg_no INT NOT NULL;  
DESC student_cgpa_auto;
```

OUTPUT:

Result Grid  Filter Rows: Export:  Wrap Cell Cont						
	Field	Type	Null	Key	Default	Extra
▶	reg_no	int	NO	PRI	NULL	
	student_name	varchar(50)	YES		NULL	
	department	varchar(20)	YES		NULL	
	no_of_subs_pass	int	YES		NULL	
	no_of_subs_fail	int	YES		NULL	
	cgpa	decimal(3,2)	YES		NULL	

RESULT: Thus, the AUTO_INCREMENT sequence was successfully created and demonstrated for generating register numbers automatically.